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CS503-Homework 1

1.1. a. $P(J|S) = \frac{P(J \cap S)}{P(S)} = \frac{0.08}{0.36} \approx 0.267$ ($P(J|S) \approx 0.267$)

b. $P(J|\bar{S}) = \frac{P(J) - P(J \cap S)}{1 - P(S)} = \frac{0.2 - 0.08}{1 - 0.3} \approx 0.171$ ($P(J|\bar{S}) \approx 0.171$)

c. $P(J \cap S | J \cup S) = \frac{P(J \cap S)}{P(J) + P(S) - P(J \cap S)} = \frac{0.08}{0.2 + 0.3 - 0.08} \approx 0.190$

($P(J \cap S | J \cup S) \approx 0.190$)

1.2. H=Harold S=Sharon

a. $P(H \cap S) = P(H) - P(H \cap \bar{S}) = 0.80 - 0.79 = 0.01$

$P(H \cup S) = P(H) + P(S) - P(H \cap S)$ ($P(H \cap \bar{S}) = 0.01$)
 $0.91 = 0.8 + 0.9 - P(H \cap S)$
 $P(H \cap S) = 0.79$

$$12.b. P(S \cap \bar{H}) = P(S) - P(H \cap S) = 0.9 - 0.79 = 0.11$$

$$P(S \cap \bar{H}) = 0.11$$

$$c. P(\bar{H} \cap \bar{S}) = 1 - P(H \cup S) = 1 - 0.91 = 0.09$$

$$P(\bar{H} \cap \bar{S}) = 0.09$$

13. J = Jerry is at the bank
S = Susan is at the bank

$$P(J \cap S) = P(J) \cdot P(S) = 0.2 \cdot 0.3 = 0.06$$

The events are not independent because the calculated $P(J \cap S)$ is not equal to the given $P(J \cap S)$.

14.a. Sum of 6 Possibilities: (1,5), (2,4), (3,3), (4,2), (5,1) $\Rightarrow 5/36$

D_1 = The sum is 6

D_2 = The second die shows 6

$$P(D_1 \cap D_2) = 1/36 \leftarrow \text{only } (1,5) \text{ works} \quad 1/36 \neq 5/216$$

$$P(D_1 \cap D_2) = P(D_1) \cdot P(D_2) = 5/36 \cdot 1/36 = 5/216$$

The two events (D_1 and D_2) are not independent.

1.4 b. D_1 = the sum is 7
 D_2 = the first dice is 5

Sum of 7 Possibilities: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) $\Rightarrow 6/36 = 1/6$

$$P(D_1 \cap D_2) = 1/36 \text{ (only (5,2) works)} \quad 1/36 = 1/36$$

$$P(D_1 \cap D_2) = 1/6 \cdot 1/6 = 1/36$$

The two events (D_1 and D_2) are independent.

1.5 TX = Texas, AK = Alaska, NJ = New Jersey, I = Oil

$$1. P(I) = P(I|TX) \cdot P(TX) + P(I|AK) \cdot P(AK) + P(I|NJ) \cdot P(NJ)$$

$$P(I) = 0.3 \cdot 0.6 + 0.2 \cdot 0.3 + 0.1 \cdot 0.1$$

$$P(I) = 0.25$$

$$2. P(TX|I) = \frac{P(I|TX) \cdot P(TX)}{P(I)} = \frac{0.3 \cdot 0.6}{0.25} = 0.72$$

$$P(TX|I) = 0.72$$

$$1.6.1. P(\text{Not Survived}) = \frac{\sum(\text{Not Survived})}{\text{Total}} = \frac{1400}{2201} \approx 0.677$$

$$P(\text{Not Survived}) \approx 0.677$$

$$2. P(1\text{st Class}) = \frac{\sum(1\text{st Class})}{\text{Total}} = \frac{329}{2201} \approx 0.148$$

$$P(1\text{st Class}) \approx 0.148$$

$$3. P(1\text{st Class} | \text{Survived}) = \frac{1\text{st Class} \cap \text{Survived}}{\sum(\text{Survived})} = \frac{203}{711} \approx 0.285$$

$$P(1\text{st Class} | \text{Survived}) \approx 0.285$$

$$4. P(\text{Survived}) = \frac{711}{2201} \approx 0.323$$

$$P(1\text{st Class}) \cdot P(\text{Survived}) = 0.148 \cdot 0.323 \approx 0.048$$

$$P(1\text{st Class} \cap \text{Survived}) = \frac{203}{2201} \approx 0.092 \quad 0.092 \neq 0.048$$

The events are not independent.

$$5. P(1\text{st Class and Child} | \text{Survived}) = \frac{1\text{st Class Child} \cap \text{Survived}}{\sum(\text{Survived})}$$

$$P(1\text{st Class Child} | \text{Survived}) = 0.008 = \frac{6}{711} = 0.008$$

$$16. b. P(\text{Adult} | \text{Survived}) = \frac{\text{Adults Survived}}{\Sigma(\text{Survived})} = \frac{654}{711} \approx 0.920$$

$$(P(\text{Adults} | \text{Survived}) \approx 0.920)$$

$$7. P(\text{Adult} | \text{1st Class} | \text{Survived}) = \frac{197}{711} \approx 0.277$$

$$P(\text{Adult} | \text{Survived}) \cdot P(\text{1st Class} | \text{Survived}) = 0.92 \cdot 0.285 \approx 0.262$$

$$0.277 \neq 0.262$$

The two events are not independent.

1.7. TP = True Positive, TN = True Negative, FP = False Positives, FN = False Negatives

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} = \frac{970 + 930}{2000} = \frac{1900}{2000} = 0.95 \quad (\text{Accuracy} = 0.95)$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{970}{970 + 70} \approx 0.933 \quad (\text{Precision} = 0.933)$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{970}{970 + 30} = 0.97 \quad (\text{Recall} = 0.97)$$

$$F1 = 2 \cdot \left(\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \right) = 2 \cdot \frac{0.933 \cdot 0.97}{0.933 + 0.97} \approx 0.951 \quad (F1 = 0.951)$$